

MATH 3220-002 PSet 9 Specification

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Subject to Change; Last Updated: 2025-11-07 16:38:56-07:00

1 Background

This problem set focuses on integral calculus. More specifically, throughout this problem set you will be investigating unsigned Riemann integrals in the context of multivariable calculus. The problems focus more on familiar ideas like the "center of mass" of an object with a mass density and the "average" of a function, as opposed to the intricacies of **Jordan content** and Jordan contentable subsets.

Barycenter Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset, $f \in R(S; \mathbb{R}_{\geq 0})$ be with positive integral. The **barycenter** (of **center of mass**) $\mathbb{B}(S, f)$ of (S, f) is defined by

$$\mathbb{B}(S, f) = \frac{\int_S f(x)xdx}{\int_S f(x)dx} \in V.$$

Here, one can consider f to be a **mass density**. $\mathbb{B}(S, 1)$ is called the barycenter of S also.

Convex Hull Let V be a normed space. Recall that a subset C of V is called **convex** if for any $x, y \in C$, the line segment connecting x and y is completely within C . Let S be an arbitrary subset of V . The **convex hull** of S is by definition the smallest convex subset of V containing S .

Mean Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset with positive Jordan content, W be a finite dimensional normed space, $f \in R(S; W)$. The **mean** (or **average**) $\mathbb{M}(S, f)$ of f on S is defined by

$$\mathbb{M}(S, f) = \frac{1}{\text{jor}(S)} \int_S f(x) dx \in W.$$

Frobenius Inner Product Let n and m be positive integers and consider the vector space $\mathcal{M}(m \times n; \mathbb{R})$ of all $m \times n$ matrices with real entries. Then the **Frobenius (or Hilbert-Schmidt) inner product** on $\mathcal{M}(m \times n; \mathbb{R})$ is defined by

$$(A, B) \mapsto \langle A, B \rangle_{\text{Fro}} = \text{tr} \left(A^T B \right).$$

We denote the associated **Frobenius norm** by

$$|A|_{\text{Fro}} = \sqrt{\langle A, A \rangle_{\text{Fro}}}.$$

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details. The **hyperlinks** too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this

problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook; it's up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. **There is nothing to prove in this problem.** Let V be an n -dimensional inner product space. Consider the following classes $\mathcal{C}$ of subsets of V :

$$\text{I: } \{V[x] \leq R \mid x \in V, R \in \mathbb{R}_{\geq 0}\}$$

(In words, this is the collection of closed balls in V .)

$$\text{II: } \left\{ \mathcal{M}_{\beta}^{-1}(I_1 \times \cdots \times I_n) \mid I_1, \dots, I_n \text{ are compact intervals, } \beta \text{ is a basis of } V, \right\}.$$

(In words, this is the collection of aligned boxes in V .)

For each $\mathcal{C}$, perform the following tasks:

- (a) For any $S \in \mathcal{C}$, compute the outer Jordan content of S as a function of parameters that define S .
- (b) For any $S \in \mathcal{C}$, compute the inner Jordan content of S as a function of parameters that define S .

- (c) For any $S \in \mathcal{C}$, compute the barycenter of S as a function of parameters that define S .
- (d) For any $S \in \mathcal{C}$, compute the outer Jordan content of $\text{bdry}(S)$ as a function of parameters that define S .
- (e) For any $S \in \mathcal{C}$, compute the inner Jordan content of $\text{bdry}(S)$ as a function of parameters that define S .
- (f) For any $S \in \mathcal{C}$, compute the barycenter of $\text{bdry}(S)$ as a function of parameters that define S .

2. **There is nothing to prove in this problem.** Let V be the space of 2×2 matrices, endowed with the Frobenius inner product. Let $S = \{X \in V \mid |X| \leq 1\}$ and $f : V \rightarrow V, X \mapsto X^2$.

Perform the following tasks:

- (a) Compute the Jordan content of S .
- (b) Compute the integral $\int_S f(X) dX$.
- (c) Compute the average $\mathbb{M}(S, 1)$.
- (d) Compute the average $\mathbb{M}(S, f)$.
- (e) Compute the barycenter $\mathbb{B}(S, 1)$.
- (f) Compute the barycenter $\mathbb{B}(S, f)$.

3. Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset with positive Jordan content.

- (a) S is convex iff $\mathbb{B}(S, 1) \in S$.
- (b) For any $f \in R(S; \mathbb{R}_{\geq 0})$ with positive integral, $\mathbb{B}(S, f)$ is in the closure of the convex hull of S .
- (c) For any point x in the closure of the convex hull of S , there is an $f \in R(S; \mathbb{R}_{\geq 0})$ with positive integral such that $\mathbb{B}(S, f) = x$.

- (d) Let $A : V \rightarrow V$ be a linear isomorphism, $b \in V$ and define $\Phi : V \rightarrow W$, $x \mapsto Ax + b$. Then for any $g \in R(\Phi(S); \mathbb{R}_{\geq 0})$ with positive integral:

$$\Phi(\mathbb{B}(S, g \circ \Phi)) = \mathbb{B}(\Phi(S), g).$$

4. Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset with positive Jordan content, W be a finite dimensional normed space, $f \in R(S; W)$. Suppose S is compact and path-connected, and f is continuous.
- (a) There is an $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$.
- (b) There is an $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$, if $\dim(W) = 1$.

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a **generative AI tool** or a **computer algebra system** for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or

- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., [ChatGPT](#)) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the [Wayback Machine](#) (see their "[Save Page Now](#)" documentation) to create a permanent, timestamped archive of your chat. There are also third-party browser extensions that are designed to extract chat logs from generative tools.
- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as [prompt engineering](#), and it is your responsibility you

design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

Here are some guides regarding prompt engineering:

- <https://platform.openai.com/docs/guides/prompt-engineering>
- <https://developers.google.com/machine-learning/resources/prompt-eng>
- <https://www.ibm.com/topics/prompt-engineering>
- <https://aws.amazon.com/what-is/prompt-engineering/>

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/SgZeggrGrUmZ38fT8>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

<https://www.gradescope.com/courses/1073533/assignments/6451791>,

see the Gradescope [documentation](#) for instructions.

You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

This problem set is due on November 14, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.